extended objects are actually part of the theory has led Paul Townsend to declare a “democracy of branes.”

Notwithstanding brane democracy, strings—one-dimensional extended objects—are special for the following reason. Physicists have shown that the mass of the extended objects of every dimension except for one-dimensional strings is inversely proportional to the value of the associated string coupling constant when we are in any of the five string regions of Figure 12.11. This means that with weak string coupling, in any of the five formulations, all but the strings will be enormously massive—orders of magnitude heavier than the Planck mass. Because they are so heavy and, therefore, from $E = mc^2$, require such unimaginably high energy to be produced, branes have only a small effect on much of physics (but not on all, as we shall see in the next chapter). However, when we venture outside the peninsular regions of Figure 12.11, the higher-dimensional branes become lighter and hence increasingly important.14

And so, the image you should have in mind is the following. In the central region of Figure 12.11, we have a theory whose fundamental ingredients are not just strings or membranes, but rather “branes” of a variety of dimensions, all more or less on equal footing. Currently, we do not have a firm grasp on many essential features of this full theory. But one thing we do know is that as we move from the central region to any of the peninsular regions, only the strings (or membranes curled up to look ever more like strings, as in Figures 12.7 and 12.8) are light enough to make contact with physics as we know it—the particles of Table 1.1 and the four forces through which they interact. The perturbative analyses string theorists have made use of for close to two decades have not been refined enough to discover even the existence of the super-massive extended objects of other dimensions; strings dominated the analyses and the theory was given the far-from-democratic name of string theory. Again, in these regions of Figure 12.11 we are justified, for most considerations, in ignoring all but the strings. In essence, this is what we have done so far in this book. We see now, though, that in actuality the theory is more rich than anyone previously imagined.

Does Any of This Solve the Unanswered Questions in String Theory?